

ECON 2010 Principles of Microeconomics

Week 4 Recitation

Zechuan Guo

Email: Zechuan.Guo@Colorado.edu

Office: ECON 307

Office Hours: Mondays & Wednesdays 2:30 PM – 4:00 PM

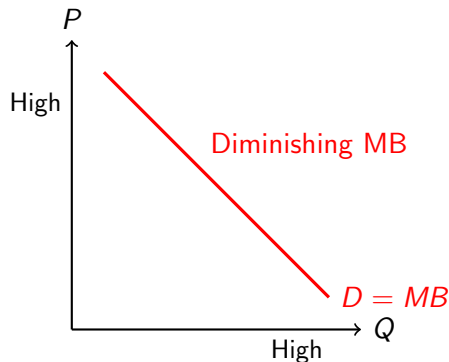
The Principles of Demand

Rational Rule for Buyers

Buy more if **Marginal Benefit** $\geq$ **Price**.

Individual demand curve $\equiv$ marginal benefit curve.

- **Downward Slopes:** Due to **diminishing marginal benefits**.
- **Market Demand:** Sum of individual demand curves
- **PEPTIC Shifters:**
 - **P**references, **E**xpectations, **P**rice of related goods, **T**ype and number of buyers, **I**ncome, **C**ongestion and network effects.



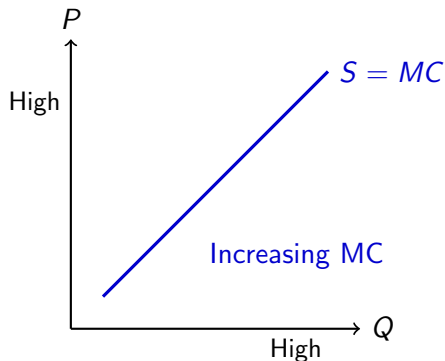
The Principles of Supply

Rational Rule for Sellers

Sell more if **Price** $\geq$ **Marginal Cost**.

Individual supply curve $\equiv$ marginal cost curve.

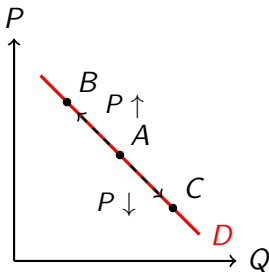
- **Upward Slopes:** Due to **increasing marginal costs**.
- **Market Supply:** Sum of individual supply curves.
- **I, POET Shifters:**
 - **I**ntermediate input prices, **P**roductivity, **O**ther output prices, **E**xpected future price, **T**ype/number of sellers.



Movements vs. Shifts in Demand

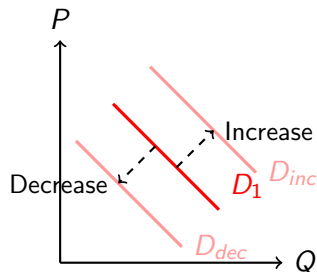
Movement Along the Curve

- Caused **only** by a change in the **price** of the good.
- Results in a change in *quantity demanded*.



Shift of the Curve

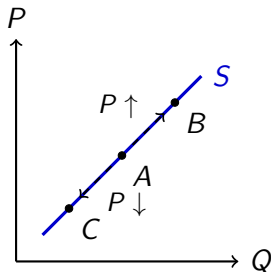
- Caused by a change in **non-price factors** (PEPTIC).
- **Increase in Demand**: Shifts **RIGHT** $\rightarrow$
- **Decrease in Demand**: Shifts **LEFT** $\leftarrow$



Movements vs. Shifts in Supply

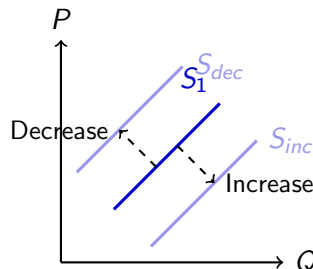
Movement Along the Curve

- Caused **only** by a change in the **price** of the good.
- Results in a change in *quantity supplied*.



Shift of the Curve

- Caused by a change in **non-price factors** (I, POET).
- **Increase in Supply**: Shifts **RIGHT** →
- **Decrease in Supply**: Shifts **LEFT** ←

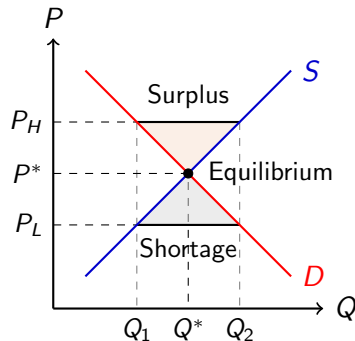


Market Equilibrium & Self-Correction

Equilibrium Condition

Point where **Quantity Supplied = Quantity Demanded**.

- **Surplus** ($P > P^*$):
 - Forces price **down**. Leads to discounting.
- **Shortage** ($P < P^*$):
 - Forces price **up**. Leads to lines/queues.



Analyzing Market Changes: The Three-Step Recipe

Step 1: Shift Source

Identify the Shifter

- Does this affect supply or demand?
- Check shift variables (**PEPTIC** or **I, POET**).

Step 2: Direction

Determine Curve Movement

- Does the curve shift **Right** (increase) or **Left** (decrease)?

Step 3: Resolve

Analyze Equilibrium

- Find the new intersection.
- Compare P^* and Q^* changes.

Shortcut Rules

Demand shifts → price & quantity move in **same direction**.

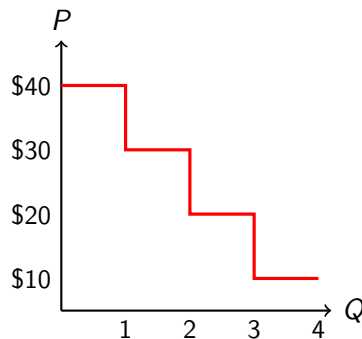
Supply shifts → price & quantity move in **opposite directions**.

Discrete Demand Curves

Buyer Benefit Thresholds

Buyers only participate if **Marginal Benefit** $\geq$ **Price**.

- **Ranking:** Plotted from highest-benefit buyer to lowest.
- **Horizontal Steps:** The exact reservation price at which an **additional buyer** is enticed to trade.
- **Vertical Portions:** Price ranges where prices change, but the quantity demanded **does not** change.

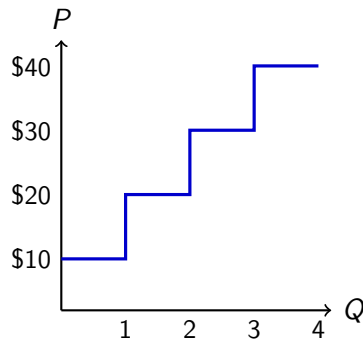


Discrete Supply Curves

Seller Cost Thresholds

Sellers only participate if **Price $\geq$ Marginal Cost**.

- **Ranking:** Plotted from lowest-cost seller to highest.
- **Horizontal Steps:** The exact reservation price at which an **additional seller** is enticed to trade.
- **Vertical Portions:** Price ranges where prices change, but the quantity supplied **does not change**.



Discrete vs. Continuous Curves

Discrete Curves

- Small number of participants.
- Clear, blocky steps.
- Visible marginal changes.

Continuous Curves

- Millions of consumers and firms.
- Microscopic steps blend together.
- Smooth, continuous lines.

Takeaway

The continuous curves represent markets with large populations. As the number of buyers and sellers increases, the incremental quantity change from one additional trader becomes mathematically insignificant relative to the total market volume, causing the discrete intervals to form a continuous line.

Worksheet Activity Instructions

- **Form Groups of 3 to 4:** Introduce yourselves to your neighbors and collaborate on the practice worksheet.
- **Navigate to Canvas:** Log in to Canvas, go to the course home page, and find the practice worksheet link under the Recitation column in the Course Schedule for Week 4.
- **Work on Problems:** Work through the problems in your groups for the next 25–30 minutes. Discuss your reasoning and help each other understand the solutions.
- **Review the Answers:** We will review the answers together as a class.

Why Theory and Reality Differ

Imperfect Information

Question: Did everyone know all the bids instantly?

Reality: In class, you had to walk around and talk to people. In real life, finding out who has the best price takes time and effort.

Negotiation & Panic

Question: How did pressure affect trading prices?

Reality: Panic, stubbornness, or aggressive bargaining can cause people to accept deals that don't match the textbook math.

Time & Effort Costs

Question: Did time limits stop some trades?

Reality: Dealing takes time. If a round ends too quickly, some buyers and sellers miss out on making a mutually beneficial deal.